

Travel Worker Timesheet and Expense Claim Management System

Mini Project Pack for Development Team (BA + Dev Handover) Updated Version - Implemented Models, Screens, Endpoints and Tests

1. Purpose

The system enables travelling workers to submit timesheets and travel-related expense claims, and enables managers to review, approve, reject, calculate, and pay those claims. The implemented solution consists of an Angular frontend, a Spring Boot REST API, JWT authentication, Spring Security authorization, and a MySQL database.

2. Scope

In scope:

- User login using JWT authentication.
- Role-based routing for employees and managers.
- Employee timesheet capture and submission.
- Expense claim capture with categories, claim details, receipts, supporting files, and bank details.
- Claim status tracking for employees.
- Manager claim review, document preview, status update, calculation, and payment processing.
- Backend storage of users, roles, timesheets, claims, claim details, claim images, bank account details, and payments.

Out of scope:

- Password reset and email notifications.
- Payroll or external banking integration.
- Audit trail beyond saved claim/payment status fields.
- Advanced reporting dashboards.

3. Users and Roles

Role	Description	Main Access
User	Travelling worker or employee submitting timesheets and claims.	Timesheet, claim submission, bank details, claim status.
Manager	Reviewer responsible for approving, rejecting, calculating, and paying claims.	Claim approval dashboard and payment action.
Admin	Seeded administrative role.	Protected admin endpoint exists, but no Angular admin screen is implemented.

Seeded users and roles are created on backend startup when missing:

- admin123 / admin@pass with role Admin
- manager123 / manager@pass with role Manager
- user / user@123 with role User

4. Functional Requirements

ID	Requirement	Implemented Behavior
FR-001	Authenticate users.	Users submit username and password to /authenticate; backend returns a JWT and user profile.

ID	Requirement	Implemented Behavior
FR-002	Store user session on frontend.	JWT and user object are saved in browser local storage.

ID	Requirement	Implemented Behavior
FR-003	Route managers to approvals.	A user with role Manager is routed to /approve-claims after login.

ID	Requirement	Implemented Behavior
FR-004	Route employees to timesheets.	Non-manager users are routed to /timesheet after login.

ID	Requirement	Implemented Behavior
FR-005	Capture timesheet rows.	Employee captures work date, location, start time, end time, and task description.

ID	Requirement	Implemented Behavior
FR-006	Calculate timesheet hours.	Frontend calculates row totals and grand total from start/end time.

ID	Requirement	Implemented Behavior
FR-007	Submit timesheets.	Completed rows are posted to /timesheets/save with JWT authorization.

ID	Requirement	Implemented Behavior
FR-008	Capture bank details.	Employee stores bank name, account number, and account type in local storage for claim submission.

ID	Requirement	Implemented Behavior
FR-009	Capture claim categories.	Employee can select Meals, Toll Fees, Other, and Distance Travelled.

ID	Requirement	Implemented Behavior
FR-010	Require claim details.	A selected claim category must have required detail fields before submission.

ID	Requirement	Implemented Behavior
FR-011	Validate meal time.	Meal claim type must match receipt time: before 12:00 Breakfast, 12:00-15:59 Lunch, 16:00+ Supper.

ID	Requirement	Implemented Behavior
FR-012	Capture distance details.	Distance Travelled requires kilometers, vehicle type, and engine size in cc.

ID	Requirement	Implemented Behavior
FR-013	Upload supporting documents.	Claim files are uploaded as multipart files and saved as ClaimImage BLOB records.

ID	Requirement	Implemented Behavior
FR-014	Save detailed receipts.	Claim detail receipt file names are matched to uploaded files and saved as BLOB data when present.

ID	Requirement	Implemented Behavior
FR-015	Generate claim reference.	Backend generates a unique 10-digit reference using claim date and current time.

ID	Requirement	Implemented Behavior
FR-016	Default claim status.	New claims default to Submitted.

ID	Requirement	Implemented Behavior
FR-017	Track employee claim status.	Employee status screen refreshes claim status every 2 seconds and on focus/navigation events.

ID	Requirement	Implemented Behavior
FR-018	List all claims for manager.	Manager dashboard loads all claims from /claims.

ID	Requirement	Implemented Behavior
FR-019	View claim documents.	Manager can request image metadata and retrieve each claim image as a blob.

ID	Requirement	Implemented Behavior
FR-020	Approve or reject claims.	Manager updates claim status through /claims/{claimId}/status.

ID	Requirement	Implemented Behavior
FR-021	Calculate claim total.	Manager calls /claims/{claimId}/calculate ; backend sums reimbursable claim detail values.

ID	Requirement	Implemented Behavior
FR-022	Display bank details during calculation.	Calculation response includes bank details attached to the claim.

ID	Requirement	Implemented Behavior
FR-023	Process claim payment.	Manager calls /claims/{claimId}/pay; backend creates a Payment and marks claim as Paid.

ID	Requirement	Implemented Behavior
FR-024	Protect API endpoints.	Most endpoints require JWT; /authenticate, /registerNewUser, GET /claims/**, and OPTIONS requests are permitted.

5. Non-Functional Requirements

ID	Requirement	Current Implementation
NFR-001	Security	JWT authentication, BCrypt password encoding, stateless Spring Security sessions.
NFR-002	Data persistence	MySQL database travelworker; Hibernate ddl-auto=update.
NFR-003	Cross-origin access	Backend allows all origins, methods GET/POST/PUT/DELETE/OPTIONS, and all headers.
NFR-004	Usability	Angular Material controls are used for forms, dialogs, lists, and date/time inputs.
NFR-005	Responsiveness	Screens include responsive CSS for smaller viewports.
NFR-006	File storage	Uploaded images and receipts are stored in database BLOB columns.
NFR-007	Availability	Local development assumes backend at http://localhost:8080 and Angular app at ng serve default port.

6. Frontend Screens

Route	Screen	Main Functions
/	Login	Capture username/password, authenticate, save JWT/user, route by role.
/timesheet	Timesheet	Capture rows, calculate hours, submit timesheet, navigate to claims/status.
/claims	Claims	Select categories, capture details, attach files, validate bank details, submit claim.
/bank-details	Bank Details	Capture and locally save bank name, account number, and account type.
/claim-status	Claim Status	Display submitted, approved, and rejected claims for the employee.
/approve-claims	Approve Claims	List all claims, view documents, calculate totals, approve/reject, process payment.

7. Data Models and Entities

7.1 Backend Entities

User

Field	Type	Notes
userId	Long	Primary key, auto generated.
userName	String	Login username.
userFirstName	String	First name.
userLastName	String	Last name.
userEmail	String	Email address.
userPassword	String	BCrypt encoded password.
role	Set<Role>	Many-to-many through USER_ROLE.
created_at	String	Created date/time string.
status	boolean	Active/inactive flag.

Role

Field	Type	Notes
roleName	String	Primary key, examples: Admin, Manager, User.
roleDescription	String	Role description.

Timesheet

Field	Type	Notes
timesheetId	Long	Primary key, auto generated.
userId	Long	References employee user.
workDate	String	Date worked, submitted as ISO string.
startTime	String	Start time/date value.
endTime	String	End time/date value.
total_hours	Double	Calculated hours for the row.
location	String	Work location.
description	String	Task description.
status	Boolean	Submission/status flag.

Claim

Field	Type	Notes
claimId	Long	Primary key, auto generated.
userId	Long	Employee user id.
claimReference	String	Generated unique reference.
claimDate	Date	Date claim was issued.
categories	String	Comma-separated selected categories.
claimImages	List<String>	Element collection of uploaded file names.
status	String	Submitted, Approved, Rejected, Paid.
total_amount	Double	Calculated total.
managerId	Long	Manager identifier if used.
userName	String	Transient display field.
userSubmissionCount	Long	Transient count field.

ClaimDetail

Field	Type	Notes
claimDetailId	Long	Primary key, auto generated.
claimId	Long	Parent claim id.
category	String	Meals, Toll Fees, Other, Distance Travelled.
detailType	String	Meal type or Distance.
description	String	Description for Other and supporting detail.
kilometers	Double	Distance travelled.
vehicleType	String	Petrol or Diesel.
engineSizeCc	Double	Engine size used for rate.
receiptTime	LocalTime	Used for meal classification.
amount	Double	Claimed amount.
allowedAmount	Double	Meal allowance reference amount.
reimbursableAmount	Double	Normalized reimbursement value.
receiptFileName	String	Uploaded receipt file name.
receiptContentType	String	Receipt MIME type.
receiptData	byte[]	Receipt BLOB.
receiptUrl	String	Transient URL for receipt download/view.

ClaimImage

Field	Type	Notes
imageId	Long	Primary key, auto generated.
claimId	Long	Parent claim id.
fileName	String	Original uploaded file name.
contentType	String	MIME type.
data	byte[]	File BLOB.

BankAccountDetails

Field	Type	Notes
bankDetailsId	Long	Primary key, auto generated.
claimId	Long	Claim id.
userId	Long	Employee user id.
bankName	String	Bank name.
accountNumber	String	Account number.
accountType	String	Cheque, Savings, or Transmission.

Payment

Field	Type	Notes
paymentId	Long	Primary key, auto generated.
claimId	Long	Paid claim id.
paymentDate	Date	Payment processing date.
reference	String	PAY- plus claim reference or claim id.
status	Boolean	True when payment processed.
processedBy	String	Manager/processor name.

7.2 API DTOs and Response Models

Model	Fields
JwtRequest	userName, userPassword
JwtResponse	user, jwtToken
ClaimImageResponse	imageId, fileName, contentType, imageUrl
ClaimCalculationResponse	claimId, claimReference, claimDate, totalAmount, details, bankDetails
UserClaimCountResponse	userSubmissionCount
PaymentResponse	paymentId, claimId, paymentDate, reference, status, processedBy

8. Business Rules

8.1 Claim Status Rules

- New claims are saved as Submitted.
- Manager may update status to Approved or Rejected.

- Payment processing changes claim status to Paid.
- Employee status screen treats missing, boolean true, or empty status as Submitted.

8.2 Meal Rules

Receipt Time	Meal Type	Allowed Amount
Before 12:00	Breakfast	R150.00
12:00 to 15:59	Lunch	R190.00
16:00 and later	Supper	R210.00

Frontend validation requires the selected meal type to match the receipt time. Backend normalizes the meal type and allowance from the submitted receipt time.

8.3 Distance Rate Rules

Petrol reimbursement rates:

Engine Size	Rate per km
<= 1250 cc	R3.172
<= 1550 cc	R3.992
<= 1750 cc	R4.337
<= 1950 cc	R4.999
<= 2150 cc	R5.359
<= 2500 cc	R6.068
<= 3500 cc	R7.576
> 3500 cc	R8.952

Diesel reimbursement rates:

Engine Size	Rate per km
<= 1250 cc	R3.197
<= 1550 cc	R3.828
<= 1750 cc	R4.248
<= 1950 cc	R4.448
<= 2150 cc	R5.196
<= 2500 cc	R5.942
> 2500 cc	R7.375

Distance amount = kilometers multiplied by the rate determined by vehicle type and engine size.

8.4 Calculation Rules

- Distance Travelled: included amount is kilometers multiplied by applicable fuel/engine rate.
- Meals: included amount is the submitted amount.
- Toll Fees: included amount is the submitted amount.
- Other: included amount is the submitted amount.
- The calculated claim total is saved to Claim.total_amount.

9. Use Cases

UC-001 Login

Actor: User, Manager, Admin

Preconditions: Account exists and is active.

Main flow:

- 1. Actor opens login screen.
- 2. Actor enters username and password.
- 3. Frontend posts credentials to /authenticate.
- 4. Backend validates credentials and returns JWT plus user profile.
- 5. Frontend stores JWT and user profile.
- 6. Manager is redirected to approvals; other users are redirected to timesheet.

UC-002 Submit Timesheet

Actor: User

Preconditions: User is logged in.

Main flow:

- 1. User opens timesheet screen.
- 2. User enters work date, location, start time, end time, and task description.
- 3. Frontend calculates total hours per row and grand total.
- 4. User submits completed rows.
- 5. Frontend posts rows to /timesheets/save.
- 6. Backend saves timesheets.
- 7. User is routed to claims screen.

UC-003 Capture Bank Details

Actor: User

Preconditions: User is logged in.

Main flow:

- 1. User opens bank details screen.
- 2. User enters bank name, account number, and account type.
- 3. Frontend validates required fields.
- 4. Frontend stores bank details locally for the next claim submission.

UC-004 Submit Claim

Actor: User

Preconditions: User is logged in and bank details are captured.

Main flow:

- 1. User opens claims screen.
- 2. User selects one or more claim categories.
- 3. User captures required category detail.
- 4. User attaches supporting documents or receipts.
- 5. Frontend validates selected categories and bank details.
- 6. Frontend posts a multipart request to /claims/save.
- 7. Backend saves claim, files, details, bank details, and generated reference.
- 8. Frontend displays confirmation with claim reference.

UC-005 Track Claim Status

Actor: User

Preconditions: User has submitted one or more claims.

Main flow:

- 1. User opens claim status screen.
- 2. Frontend refreshes user claims from `/claims/user/{userId}` and count from `/claims/user/{userId}/count`.
- 3. Screen displays submitted, approved, and rejected claims.
- 4. Screen refreshes periodically and when browser focus returns.

UC-006 Review and Decide Claim

Actor: Manager

Preconditions: Manager is logged in.

Main flow:

- 1. Manager opens approve claims screen.
- 2. Frontend loads claims from `/claims`.
- 3. Manager views claim fields and supporting documents.
- 4. Manager approves or rejects a claim.
- 5. Frontend sends status update to `/claims/{claimId}/status`.
- 6. Backend saves the new status.

UC-007 Calculate and Pay Claim

Actor: Manager

Preconditions: Claim exists and has claim details.

Main flow:

- 1. Manager selects Calculate on a claim.
- 2. Frontend calls `/claims/{claimId}/calculate`.
- 3. Backend calculates total amount and returns details plus bank details.
- 4. Manager reviews calculation and selects Pay.
- 5. Frontend posts processor name to `/claims/{claimId}/pay`.
- 6. Backend creates payment record and marks claim as Paid.

10. Suggested API Endpoints

Base URL: `http://localhost:8080`

Method	Endpoint	Purpose	Auth
POST	/authenticate	Login and receive JWT.	Public
POST	/registerNewUser	Register a new user with User role.	Public
POST	/createNewRole	Create a role.	JWT
GET	/forAdmin	Admin-only protected test endpoint.	Admin
GET	/forUser	User-only protected test endpoint.	User
POST	/timesheets/save	Save submitted timesheet rows.	JWT
GET	/timesheets	List all timesheets.	JWT
POST	/claims/save	Save claim as multipart form data.	JWT
GET	/claims	List all claims.	Public in current security config
GET	/claims/user/{userId}	List claims for one user.	Public in current security config
GET	/claims/user/{userId}/count	Count claims for one user.	Public in current security config
GET	/claims/{claimId}/images	Get claim image metadata.	Public in current security config
GET	/claims/{claimId}/images/{imageId}	Get image BLOB.	Public in current security config
GET	/claims/{claimId}/details	Get claim details.	Public in current security config
GET	/claims/{claimId}/details/{detailId}/receipt	Get claim detail receipt BLOB.	Public in current security config
GET	/claims/{claimId}/calculate	Calculate and persist claim total.	Public in current security config
PUT	/claims/{claimId}/status	Update claim status.	JWT
POST	/claims/{claimId}/pay	Create payment and mark claim Paid.	JWT

Recommended future endpoint hardening:

- Restrict GET /claims/** to authenticated users.
- Restrict manager-only actions such as status updates, calculation, and payment to role Manager.
- Add user ownership checks for employee claim status endpoints.
- Add CRUD endpoints for bank details if bank details should persist independently before claim submission.

11. ERD

Diagram source (mermaid)

```
erDiagram
    USER {
        long userId PK
        string userName
        string userFirstName
        string userLastName
        string userEmail
        string userPassword
        string created_at
        boolean status
    }

    ROLE {
        string roleName PK
        string roleDescription
    }

    USER_ROLE {
        long USER_ID FK
        string ROLE_ID FK
    }

    TIMESHEET {
        long timesheetId PK
        long userId FK
        string workDate
        string startTime
        string endTime
        double total_hours
        string location
        string description
        boolean status
    }

    CLAIM {
        long claimId PK
        long userId FK
        string claimReference
        date claimDate
        string categories
        string status
        double total_amount
        long managerId
    }

    CLAIM_DETAIL {
        long claimDetailId PK
        long claimId FK
        string category
        string detailType
        string description
        double kilometers
        string vehicleType
        double engineSizeCc
        time receiptTime
        double amount
        double allowedAmount
        double reimbursableAmount
        string receiptFileName
        string receiptContentType
        blob receiptData
    }

    CLAIM_IMAGE {
        long imageId PK
        long claimId FK
        string fileName
        string contentType
        blob data
    }
```

Diagram source (mermaid)

```
BANK_ACCOUNT_DETAILS {
  long bankDetailsId PK
  long claimId FK
  long userId FK
  string bankName
  string accountNumber
  string accountType
}

PAYMENT {
  long paymentId PK
  long claimId FK
  date paymentDate
  string reference
  boolean status
  string processedBy
}

USER ||--o{ TIMESHEET : submits
USER ||--o{ CLAIM : submits
USER ||--o{ BANK_ACCOUNT_DETAILS : owns
USER }o--o{ ROLE : has
USER_ROLE }o--|| USER : maps
USER_ROLE }o--|| ROLE : maps
CLAIM ||--o{ CLAIM_DETAIL : contains
CLAIM ||--o{ CLAIM_IMAGE : has
CLAIM ||--o{ BANK_ACCOUNT_DETAILS : uses
CLAIM ||--o{ PAYMENT : paid_by
```

12. Process Flow Diagrams

12.1 Authentication and Routing

Diagram source (mermaid)

```
flowchart TD
  A[Open Login Screen] --> B[Enter username and password]
  B --> C[POST /authenticate]
  C --> D{Valid credentials-}
  D -- No --> E[Show invalid username or password]
  D -- Yes --> F[Save JWT and user profile]
  F --> G{User has Manager role-}
  G -- Yes --> H[Go to /approve-claims]
  G -- No --> I[Refresh claim status]
  I --> J[Go to /timesheet]
```

12.2 Employee Timesheet and Claim Submission

Diagram source (mermaid)

```
flowchart TD
  A[Employee logs in] --> B[Capture timesheet rows]
  B --> C[Calculate row and grand total hours]
  C --> D{At least one complete row-}
  D -- No --> E[Show validation message]
  D -- Yes --> F[POST /timesheets/save]
  F --> G[Open claims screen]
  G --> H[Select claim categories]
  H --> I[Capture required category details]
  I --> J[Attach supporting documents]
  J --> K{Bank details available-}
  K -- No --> L[Open bank details screen]
  K -- Yes --> M[POST multipart /claims/save]
  M --> N[Backend saves claim, details, files, bank details]
  N --> O[Show claim reference]
```

12.3 Manager Approval, Calculation, and Payment

Diagram source (mermaid)

```
graph TD
    A[Manager logs in] --> B[Load all claims]
    B --> C[Review claim list]
    C --> D[View documents]
    C --> E[Approve or reject]
    E --> F[PUT /claims/{claimId}/status]
    C --> G[Calculate claim]
    G --> H[GET /claims/{claimId}/calculate]
    H --> I[Show details, bank details, total]
    I --> J[Process payment]
    J --> K[POST /claims/{claimId}/pay]
    K --> L[Create Payment and mark claim Paid]
```

13. Test Scenarios

ID	Scenario	Expected Result
TS-001	Login with valid employee credentials.	JWT and user are saved; employee is routed to /timesheet.

ID	Scenario	Expected Result
TS-002	Login with valid manager credentials.	JWT and user are saved; manager is routed to /approve-claims.

ID	Scenario	Expected Result
TS-003	Login with invalid credentials.	Error message Invalid username or password is shown.

ID	Scenario	Expected Result
TS-004	Submit timesheet without login.	Error message asks user to log in.

ID	Scenario	Expected Result
TS-005	Submit timesheet with no complete rows.	Validation message asks for at least one row.

ID	Scenario	Expected Result
TS-006	Enter end time before start time.	Row total displays Invalid; invalid total is excluded from grand total.

ID	Scenario	Expected Result
TS-007	Submit a valid timesheet row.	Backend saves row and frontend navigates to claims screen.

ID	Scenario	Expected Result
TS-008	Save blank bank details.	Validation requires bank name, account number, and account type.

ID	Scenario	Expected Result
TS-009	Save complete bank details.	Details are stored locally and success message is shown.

ID	Scenario	Expected Result
TS-010	Submit claim without category.	Validation asks user to select at least one claim category.

ID	Scenario	Expected Result
TS-011	Submit claim without bank details.	Validation asks user to capture bank details before submission.

ID	Scenario	Expected Result
TS-012	Select Meals without detail.	Detail dialog opens before submission.

ID	Scenario	Expected Result
TS-013	Select Breakfast with receipt time 13:00.	Frontend blocks save and says slip time is treated as Lunch.

ID	Scenario	Expected Result
TS-014	Submit Distance Travelled without kilometers.	Category remains incomplete and submission is blocked.

ID	Scenario	Expected Result
TS-015	Submit valid claim with files and bank details.	Backend saves claim, generated reference is returned, confirmation dialog appears.

ID	Scenario	Expected Result
TS-016	Manager loads approvals.	All claims are displayed or an error appears if unavailable.

ID	Scenario	Expected Result
TS-017	Manager views claim images.	Image metadata loads and BLOB previews are created.

ID	Scenario	Expected Result
TS-018	Manager approves a claim.	Claim status changes to Approved and persists on backend.

ID	Scenario	Expected Result
TS-019	Manager rejects a claim.	Claim status changes to Rejected and persists on backend.

ID	Scenario	Expected Result
TS-020	Employee opens status screen after approval.	Approved claim appears with approved status/icon.

ID	Scenario	Expected Result
TS-021	Calculate claim with Meals detail.	Total includes submitted meal amount.

ID	Scenario	Expected Result
TS-022	Calculate claim with Toll Fees detail.	Total includes submitted toll amount.

ID	Scenario	Expected Result
TS-023	Calculate claim with Other detail.	Total includes submitted other amount.

ID	Scenario	Expected Result
TS-024	Calculate petrol distance claim.	Total equals kilometers multiplied by petrol cc rate.

ID	Scenario	Expected Result
TS-025	Calculate diesel distance claim.	Total equals kilometers multiplied by diesel cc rate.

ID	Scenario	Expected Result
TS-026	Pay a calculated claim.	Payment is saved, reference starts with PAY-, claim status becomes Paid.

ID	Scenario	Expected Result
TS-027	Access protected POST without JWT.	Backend returns unauthorized response.

ID	Scenario	Expected Result
TS-028	Register a new user.	User is saved with encoded password and User role.

14. Assumptions and Open Items

- The previous PDF could not be extracted in this environment because no local PDF text extraction tool or Python PDF library is installed.
- The current backend permits GET access to /claims/**; this is useful during development but should be restricted for production.
- Bank details are saved in browser local storage until a claim is submitted, then persisted with that claim.
- No frontend registration screen exists although /registerNewUser exists on the backend.
- No Angular route guard is implemented; routing checks are handled inside components.
- Existing unit test files are mostly generated skeletons and should be expanded around the scenarios listed above.